

Smart Spring Monitoring for Kisii & Nyamira: Real-Time Water Security & Climate Resilience

Funding Proposal

Submitted by:

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Requested Funding Amount: KES 3,500,000 (USD 25,900)

Proposed Project Duration: 18 Months (October 2025 – March 2027)

Target Donors: World Bank (KEWASIP), Green Climate Fund, Adaptation Fund, IFAD, County Climate Change Funds, African Water Facility

1 Executive Summary

Kisii and Nyamira counties, with over 2 million residents, face water security challenges, including pollution from agricultural runoff, declining spring flows due to climate variability, and inadequate real-time monitoring, exacerbating health risks and flood vulnerabilities. This student-led project proposes a digital twin system for key rivers—Gucha (Kuja), Mogonga, Sondu, Eaka, Kijauri, Kemera, Charachani, Bisembe, Chirichiro, Ramacha, and streams like Mogusii, Riana, and Iyabe—with real-time pollution detection using IoT sensors to monitor chemical composition (pH, EC, turbidity, DO, temperature), spring discharge, water speed, and levels. Sensors will be deployed at 50 points, including every 100 meters in Kisii town sections and at stream confluences, with at least 5 springs per river. Aligned with Kenya's National Climate Change Action Plan, SDGs 6 and 13, and inspired by IWMI's digital twin prototypes, it enhances resilience for upstream and downstream communities.

Implemented in five phases over 18 months, outcomes include a digital twin, real-time alerts reducing response times, and trained communities. With a budget of KES 3,500,000 (45% for sensors), it leverages JKUAT support and partnerships. Funding will empower women and youth, addressing gender inequalities in water access.

2 Background and Problem Statement

Kisii and Nyamira, with populations exceeding 1.2 million and 600,000, rely on springs and rivers like Gucha (Kuja), Mogonga, Sondu, Eaka, Kijauri, Kemera, Charachani, Bisembe, Chirichiro, Ramacha, and streams (Mogusii, Riana, Iyabe). Piped water coverage is minimal (<30% urban), forcing use of untreated sources. Challenges include:

- **Pollution:** Runoff contaminates springs/rivers, causing diseases like cholera.
- **Climate Variability:** Rainy seasons cause floods; declining flows threaten scarcity.
- **Urban/Confluence Risks:** Dense pollution in Kisii town; confluences amplify downstream impacts.
- **No Monitoring:** Lack of real-time systems delays responses.

These deepen insecurity, burdening women and girls who fetch water.

3 Project Objectives

1. Map springs and rivers, selecting 5+ springs per river for monitoring.
2. Deploy sensors at 50 points for real-time pollution (pH/EC/turbidity/DO/temp) and hydrological parameters (discharge, speed, level).
3. Develop a digital twin integrating data for simulations.

4. Create dashboards/alerts for early warnings.
5. Build capacity for sustainable, gender-inclusive management.

4 Methodology

Adopting a phased approach inspired by IWMI and Adaptation Fund projects, using QGIS, Python, and IoT probes (e.g., Tuya WiFi multiparameter).

- **Phase 1: Mapping (Months 1–3):** GPS survey of rivers/springs; select 50 points, including 5+ springs/river, Kisii town (every 100m), confluences.
- **Phase 2: Assessment (Months 4–6):** Manual tests; lab analysis (50 samples).
- **Phase 3: Deployment (Months 7–9):** Install 50 solar-powered IoT sensors; GSM/LoRa.
- **Phase 4: Digital Twin (Months 10–12):** Cloud model with alerts (e.g., pH drop triggers SMS).
- **Phase 5: Engagement (Months 13–18):** Train 200+ stakeholders; integrate systems.

Inclusivity: Women-led training; data privacy compliance.

5 Expected Outcomes

- Digital twin simulating pollution and hydrology.
- Real-time detection at 50 points, reducing response times.
- Alerts averting risks for 100,000+ beneficiaries.
- Policy data for adaptation plans.
- Resilience benefits, e.g., reduced water collection time for women.

6 Detailed Budget

Total: KES 3,500,000. Optimized for 50 sensor points using affordable IoT kits (e.g., Tuya WiFi KES 15,000/unit for pollution; low-cost loggers KES 5,000; solar KES 5,000). Sources: Jumia, Hanna, Cropnuts.

6.1 Field Team Costs (8 Trips, Team of 4)

Assumes 10 days/trip for expanded mapping.

6.2 Field Equipment (One-Time)

Reusable; borrow from JKUAT.

Item	Details	Cost/Trip (KES)	Trips	Total (KES)	Justification
Long-distance transport	Van hire (14-seater)	30,000	8	240,000	Coverage of Kaya
Local transport	Motorbikes	12,000	8	96,000	For 5
Accommodation	4 people × 10 nights × 1,500	60,000	8	480,000	Safety
Meals	4 people × 500/day × 10	20,000	8	160,000	Govt.
Local allowances	Guides	15,000	8	120,000	Comm
Subtotal				1,096,000	Partn 20%.

Item	Details	Cost (KES)	Justification & Sources
Handheld GPS	Garmin eTrex	25,000	Mapping; Garmin ~KES 27,000.
Multiparameter probes	Portable	150,000	Hanna ~KES 150,000.
Turbidity meter	Hanna	50,000	Complements.
Flow meter	Portable	70,000	Discharge.
Sample bottles/PPE	Lab/safety	25,000	Reusable.
Data devices	Smartphones/ODK	20,000	Surveys.
Subtotal		340,000	Borrow to save.

6.3 Real-Time Monitoring Sensors

50 points: Combined kits (pH/EC/turbidity/DO/temp/speed + flow/level) ~KES 20,000/unit; solar ~KES 5,000.

Item	Details	Unit Cost (KES)	Qty	Total (KES)	Justification
Multiparameter IoT kits	pH/EC/turbidity/DO/temp/speed	15,000	50	750,000	Pollution 15k.
Flow/level loggers	Discharge/velocity	5,000	50	250,000	Hydrology
Data gateway/GSM	Connectivity (annual)	30,000	1	30,000	Dashboard
Solar kits	Off-grid	5,000	50	250,000	Sustainability
Subtotal				1,280,000	Scalability

6.4 Consumables & Lab Analysis

For 50 samples.

6.5 Digital Twin System & Documentation

Open-source focus.

6.6 Contingency (10%)

On KES 3,011,000: KES 301,100.

Item	Details	Cost (KES)	Justification & Sources
Lab analysis	Nutrients/metals; 50 samples @ KES 3,500	175,000	Baseline; Cropnuts.
Field consumables	Filters/reagents	30,000	Disposable.
Subtotal		205,000	JKUAT discounts.

Item	Details	Cost (KES)	Justification & Sources
Printing/stationery	Reports	10,000	Deliverables.
Internet/cloud	Annual hosting	30,000	Real-time.
Software	QGIS/ArcGIS optional	20,000	Simulations.
Training materials	Workshops for 200+	30,000	Sustainability.
Subtotal		90,000	Accessibility.

6.7 Grand Total

KES 3,500,000 (USD 25,900). Grant-ready: ROI via alerts; SDG alignments. Procurement: Jumia/Hanna; partnerships save 15%.

7 Timeline

- Q4 2025–Q1 2026: Phases 1–2 (Mapping/Assessment).
- Q2–Q3 2026: Phase 3 (Deployment).
- Q4 2026: Phase 4 (Digital Twin).
- Q1 2027: Phase 5 (Engagement); reporting.

8 Partners and Stakeholders

- **Lead:** JKUAT Department of Computer Technology.
- **Local:** Kisii/Nyamira County Governments, Water Resources Authority, water committees.
- **Community:** Farmers, women's groups.
- **Collaborators:** IWMI for digital twin expertise.

9 Monitoring and Evaluation

Quarterly KPIs (e.g., 50 sensors deployed, trainings held, alerts issued). JKUAT audit; community surveys. Alignment with Adaptation Fund metrics.

10 Sustainability

Solar-powered sensors; trained committees; county system integration. Expand via KEWASIP/GCF.

11 Risks and Mitigation

- **Connectivity:** LoRa fallback; Safaricom partnership.
- **Vandalism:** Community ownership; insurance in contingency.
- **Overruns:** 10% buffer; JKUAT cost-sharing.
- **Delays:** Flexible scheduling.

12 Conclusion and Funding Request

This project uses sensors for real-time pollution and hydrological monitoring across 50 points on key rivers, addressing gaps in Kisii and Nyamira. We request KES 3,500,000 for a pilot with scaling potential. Support local innovation for Kenya's climate goals. Contact for details.